AERIS-10 - STM32F746 firmware layered architecture (hand-drawn from source)
Drawing ID: SD-03 Revision: A Date: 2026-10-09 Status: PARTIAL - HAL/CubeMX/USB-middleware layer is absent from the repository (dashed red)
Source: 9_Firmware/9_1_Microcontroller/9_1_3_C_Cpp_Code/main.cpp (2411 lines), main.h; 9_1_1_C_Cpp_Libraries/*.cl*.cppl*.h (131 files: ADI no-OS, ST HAL glue, sensor drivers); docs/STM32/STM32_PROJECT_RECONSTRUCTION.md
CODE = 9_1_3_C_Cpp_Code, LIB = 9_1_1_C_Cpp_Libraries. Every edge is an #include or a call found in the sources (file:line in the label).

MX_*_Init call HAL_*_Init -> HAL_*_MspInit (stm32f7xx_hal_msp.c); stm32f7xx_hal_conf.h selects HAL modules
HAL / CubeMX GENERATED LAYER
PRESENT in LIB/: stm32f7xx_hal_conf.h (modules enabled :38-88, HSE_VALUE 25 MHz :97) stm32f7xx_hal_msp.c (I2C :86-190, SPI :277-324, UART :433-507 pin/AF setup) stm32f7xx_it.c (OTG_FS_IRQHandler :204-209 -> HAL_PCD_IRQHandler(&hpcd_USB_OTG_FS)) system_stm32f7xx.c, syscalls.c, sysmem.c, errno.h
stm32f7xx_hal_conf.h includes stm32f7xx_hal_*.h; it.c :58 extern hpcd_USB_OTG_FS (defined in missing usbd_conf.c)
MISSING from repository: stm32f7xx_hal.h + STM32F7xx_HAL_Driver sources (STM32CubeF7 package) CMSIS device headers (stm32f746xx.h), startup_stm32f746xx.s, linker script (*.ld) usb_device.c/.h, usbd_cdc_if.c/.h, usbd_desc.c/.h, usbd_conf.c/.h (defines hpcd_USB_OTG_FS, PA11/PA12 AF10) STM32_USB_Device_Library Core + Class/CDC .ioc CubeMX project
#include stm32f7xx_hal.h (main.h:30, main.cpp:68); HAL_Init, HAL_GPIO_*, HAL_Delay, MX_USB_DEVICE_Init
#include stm32f7xx_hal.h (.h:5); HAL_SPI_*, HAL_GPIO_*, HAL_UART_*
#include stm32f7xx_hal.h (.H:8); HAL_I2C_*
#include stm32f7xx_hal.h (.H:8); HAL_I2C_*
#include stm32f7xx_hal.h (.h:50); #include stm32f7xx_hal.h (.h:4); HAL_I2C_*
#include usb_device.h, usbd_cdc_if.h (gps_handler.h:6-7); CDC_Transmit_FS :118
#include stm32f7xx_hal.h (stm32_spi.h:5); HAL_SPI_TransmitReceive, HAL_UART_*
external devices (schematic nets per docs/STM32/STM32_PROJECT_RECONSTRUCTION.md section 3.2-3.3)
4x ADAR1000 via FPGA level shift (bank 34, 1.8 V) RTL pass-through undriven - see SD-02
2x DAC5578 @0x48/0x49 (PA gate bias)
FPGA XC7A50T DIG_0..4 (F13/E16/D16/F15/E15, PIN_MAP:16-20)
3x ADS7830 @0x48/0x4A (Idq), @0x49 (8x TMP37)
GPS module (JP8, NMEA 9600)
GY-85 IMU + BMP180 (JP7/JP18)
Frequency Synthesizer board: AD9523 (JP1), ADF4382 TX/RX (JP13<->JP2)
Host PC (mini-USB X53, CDC)
PERIPHERALS (handles CODE/main.cpp:108-118; pins LIB/stm32f7xx_hal_msp.c; GPIO CODE/main.h:62-165)
HAL_UART_Transmit(&huart3) :48,60,68
USART3 huart3 PB10 / PB11 AF7 (msp:506-507) MX_USART3_UART_Init :2178 (115200, debug text)
SPI1 hspi1 PA5/PA6/PA7 AF5 (msp:298-299) MX_SPI1_Init :2016, prescaler /2 CS PA0..PA3 (main.h:78-85)
&hspi1 (ADAR1000_Manager.cpp:8,619)
I2C1 hi2c1 PB6 SCL / PB7 SDA AF4 (msp:114-115) MX_I2C1_Init :1872
TIM1 htim1 MX_TIM1_Init :2096-2135 (PSC 71) HAL_TIM_Base_Start :1232
HAL_GPIO_TogglePin(GPIOD, PIN_8/9/10) :448,484,514
hdac->hi2c = &hi2c1 (main.cpp:1563,1570)
GPIO (MX_GPIO_Init :2213-2330) PD8..PD12 outputs -> FPGA DIG_0..4 (:2306-2311) PD13..PD15 inputs DIG_5..7 (:2313-2317, never read) EN_P_* rail enables PE7..PE15, PG0..PG5; AD9523 ctrl PF3..PF10; ADF4382 ctrl PG6..PG15; DAC CLR/LDAC PB4/5/8/9; stepper PD4/PD5; PD6 RFPA VDD; PD7 cooling; LEDs PF12..PF15
&hi2c2 (main.cpp:1605,1611,1670)
I2C2 hi2c2 PF0 / PF1 AF4 (msp:147-148) MX_I2C2_Init :1920
UART5 huart5 PC12 TX / PD2 RX AF8 (msp:465-473) MX_UART5_Init :2143 (9600)
HAL_UART_Receive(&huart5) :908
I2C3 hi2c3 PA8 SCL / PC9 SDA AF4 (msp:181-189) MX_I2C3_Init :1968
SPI4 hspi4 PE2/PE5/PE6 AF5 (msp:323-324) MX_SPI4_Init :2056 CS: AD9523 PF7 (main.h:70), ADF4382 TX PG14 / RX PG10 (main.h:150,142)
USB OTG_FS hpcd_USB_OTG_FS MX_USB_DEVICE_Init :1229 (definition missing) PA11/PA12 - from schematic net STM32_USB_FS_D_N/D_P, AF setup lives in missing usbd_conf.c -> UNVERIFIED in repo
extra=&hspi4 (main.cpp:1040; adf4382a_manager.c:43,52)
#include usb_device.h :21, usbd_cdc_if.h :23; USBD_CDC_SetRxBuffer/ReceivePacket :336-337; CDC_Transmit_FS :1692
DRIVERS (LIB/)
ADAR1000_Manager.cpp/.h (C++) extern hspi1, huart3 (:8-9); HAL_SPI_Transmit(Receive)(&hspi1) :619-621 CHIP_SELECTS PA0..PA3 :15-20, :629; rail GPIO EN_P_* :51-79, 332-370 pulseTXMode/pulseRXMode/setCustomBeamPattern16 (.h:53-59)
pulseTXMode/pulseRXMode :449-451, setCustomBeamPattern16 :486-487
DA5578.c / DAC5578.H HAL_I2C_Master_Transmit(hdac->hi2c) :69..368; LDAC/CLR GPIO :37-41 includes stm32f7xx_hal.h (.H:8)
ADS7830.c / ADS7830.H HAL_I2C_Master_Transmit/Receive(hadc->hi2c) :111-121, 180-190 includes stm32f7xx_hal.h (.H:8)
adf4382a_manager.c/.h includes main.h, adf4382.h, no_os_spi.h (.h:4-6) spi_tx/rx_param.platform_ops = NULL (:42, :51) -> no_os_spi_init fails (DEFECT C3) pin macros PG0..PG9 collide with rail enables (DEFECT C2, STM32 doc STM-T04)
adar1000.c/.h (ADI no-OS driver) included by main.cpp:24; no adar1000_* call site found in main.cpp or ADAR1000_Manager.cpp -> UNVERIFIED whether used
BMP180.cpp/.h HAL_I2C_Master_Transmit/Receive(&hi2c3) :405-409 includes stm32f7xx_hal.h (.h:50)
&hi2c3 (BMP180.cpp:405)
GY_85_HAL.c/.h (ADXL345, HMC5883, ITG3200) HAL_I2C_Master_Transmit/Receive(&hi2c3) :35-116 includes stm32f7xx_hal.h (.h:4)
&hi2c3 (GY_85_HAL.c:35)
platform_ops = NULL (:42,:51) - NOT wired to stm32_spi_ops
#include adf4382.h (adf4382a_manager.h:5)
adf4382.c/.h (ADI no-OS PLL driver) includes no_os_spi.h, no_os_util.h, no_os_units.h (adf4382.h:39-41)
BMP180 myBMP(BMP180_ULTRAHIGHRES) :166
ad9523.c/.h (ADI no-OS clock driver) includes no_os_spi.h, no_os_delay.h (ad9523.h:38-39)
gps_handler.cpp/.h + TinyGPS++.cpp / TinyGPSPlus.h gps_handler.h includes main.h, usb_device.h, usbd_cdc_if.h (:5-7) GPS_SendToGUI -> HAL_UART_Transmit(gui_huart) :60; GPS_SendBinaryToGUI -> CDC_Transmit_FS :118 GPS_Init() never called -> gui_huart == NULL guard :65-67 (DEFECT C6)
#include no_os_spi.h
#include no_os_spi.h / no_os_delay.h
ADI no-OS core (LIB/no_os_*.c/.h, 60 files) no_os_spi.c, no_os_gpio.c, no_os_delay.h, no_os_alloc.c, no_os_util.c, ... included by main.cpp:29-37
platform_ops dispatch (no_os_spi.c -> stm32_spi_ops)
no-OS STM32 platform glue (LIB/) stm32_spi.c/.h (HAL_SPI_TransmitReceive :34; chip_select ignored :23-38), stm32_gpio.c, stm32_delay.c, stm32_uart.c (HAL_UART_* :49-276), stm32_i2c.c, stm32_timer.c, stm32_irq.c, stm32_gpio_irq.c, stm32_dma.c, stm32_pwm.c, platform_noos_stm32.c/.H
USBHandler.cpp/.h + RadarSettings.cpp/.h processUSBData :15-34, processStartFlag :36-57, processSettingsData :59-93 RadarSettings::parseFromUSB (RadarSettings.h:12)
via USBD stack (missing)
APPLICATION (CODE/main.cpp, main.h)
#include adar1000.h (:24)
adarManager.initializeADTR1107Sequence/initializeAllDevices/performSystemCalibration (main.cpp:346-365)
initializeBeamMatrices() :412-443 beam-steering loop :480-516 -> executeChirpSequence() :445-470 PD8 toggle = new chirp :448,459; PD9 = elevation :484; PD10 = azimuth :514
systemPowerUpSequence() :341-410 ADTR1107 sequence, ADAR1000 init, calibration via adarManager
DAC5578_Init/SetClearCode/SetupLDAC/WriteAndUpdateChannelValue :1563-1652
ADS7830_Init/Measure_SingleEnded :1605-1653
ADS7830_Measure_SingleEnded(&hadc3) :840-847, 1753
ADF4382A_* :689, 831, 1406-1445
__HAL_TIM_GET/SET_COUNTER(&htim1) :284,293-294
PD4/PD5 :1514-1521
GY85_Init/GY85_Update :623, 1280-1282
PA bias / Idq loop :1563-1657 DAC5578_Init(&hdac1,&hi2c1,0x48) :1563, (&hdac2,&hi2c1,0x49) :1570 ADS7830_Init(&hadc1,&hi2c2,0x48) :1605, (&hadc2,&hi2c2,0x4A) :1611 EN_DIS_RFPA_VDD PD6 :1601
main() CODE/main.cpp:1200-1800 HAL_Init :1202, SystemClock_Config :1209 (HSE 25 MHz, PLL M25/N144/P2/Q3 :1821-1827) MX_GPIO/TIM1/I2C1/I2C2/I2C3/SPI1/SPI4/UART5/USART3_Init, MX_USB_DEVICE_Init :1220-1229 180 s OCXO warm-up :1237; rail enables :1241-1275, 1485-1489; FPGA reset PD12 :1659-1662
temperature monitor :840-847, 1670-1753 ADS7830_Init(&hadc3,&hi2c2,0x49) :1670; 8 channels x 0.64705
ADF4382 LO control via ADF4382A_Manager lo_manager :214 ADF4382A_Manager_Init :689, SetPhaseShift :1406, StrobePhaseShift :1409-1410, CheckLockStatus :831/1428, TriggerTimedSync :1445
delay_us() :292-295 (TIM1 counter, PSC 71 :2110) delay_ns() :300-312 (DWT->CYCCNT)
stepper bit-bang (no driver file) STEPPER_CW_P PD4, STEPPER_CLK_P PD5 (main.h:134-137) :517-521, 1514-1521 with delay_us()
sensors: GY85_Init :1280, GY85_Update :623/1282; BMP180 myBMP :166 GPS: HAL_UART_Receive(&huart5) :908 -> gps.encode() :909 (TinyGPSPlus gps :122) gps.location.lat/lng :1509-1510; GPS_SendBinaryToGUI :1531
gps.encode :909; GPS_SendBinaryToGUI :1531
ad9523_init/ad9523_setup :1046-1058
AD9523 configuration :927-1058 struct ad9523_dev, init_param.spi_init.platform_ops=&stm32_spi_ops :1039, extra=&hspi4 :1040 ad9523_init :1046, ad9523_setup :1047/1058
&stm32_spi_ops, &hspi4 :1039-1040
USB CDC application USBHandler usbHandler :128; CDC_Receive_FS() :331-338 (extern "C", never bound - see SD-04) wait loop :1534-1555; CDC_Transmit_FS(initial_status) :1692
usbHandler.processUSBData :333; isStartFlagReceived/getState/getSettings :1535-1538

LEGEND / EVIDENCE
blue = application code; green cluster = driver sources present in LIB/; purple = HAL layer (present vs MISSING);
orange = present but with a documented defect; red dashed = absent from repository; grey dotted = included but no call site found (UNVERIFIED)
solid edge = #include or direct call (file:line); dotted edge = HAL handle passed/used; dashed red edge = depends on a missing file
Handles: CODE/main.cpp:108-118 (hi2c1..3, hspi1, hspi4, htim1, huart5, huart3), :129 extern hUsbDeviceFS
Init order: CODE/main.cpp:1202-1229
HAL modules enabled: LIB/stm32f7xx_hal_conf.h:38-88 (HAL, SPI, TIM, UART, PCD, GPIO, EXTI, DMA, RCC, FLASH, PWR, I2C, CORTEX); no DMA handle is ever created
Known firmware defects C1-C7: docs/STM32/STM32_PROJECT_RECONSTRUCTION.md section STM-T04
NOT VERIFIED: compilation (no toolchain/HAL in repo), USB VID/PID, interrupt priorities, I2C bus speeds, whether adar1000.c is linked